

2018 ICAS Science Paper B

1. All insects have six legs and one pair of antennae on their heads.

Which of these is **NOT** an insect?

(A)



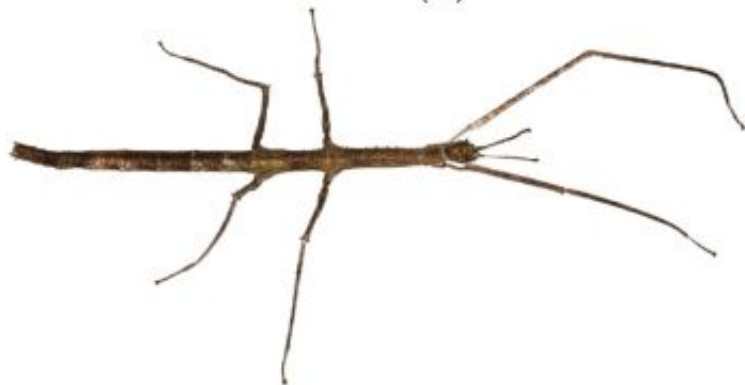
(B)



(C)



(D)



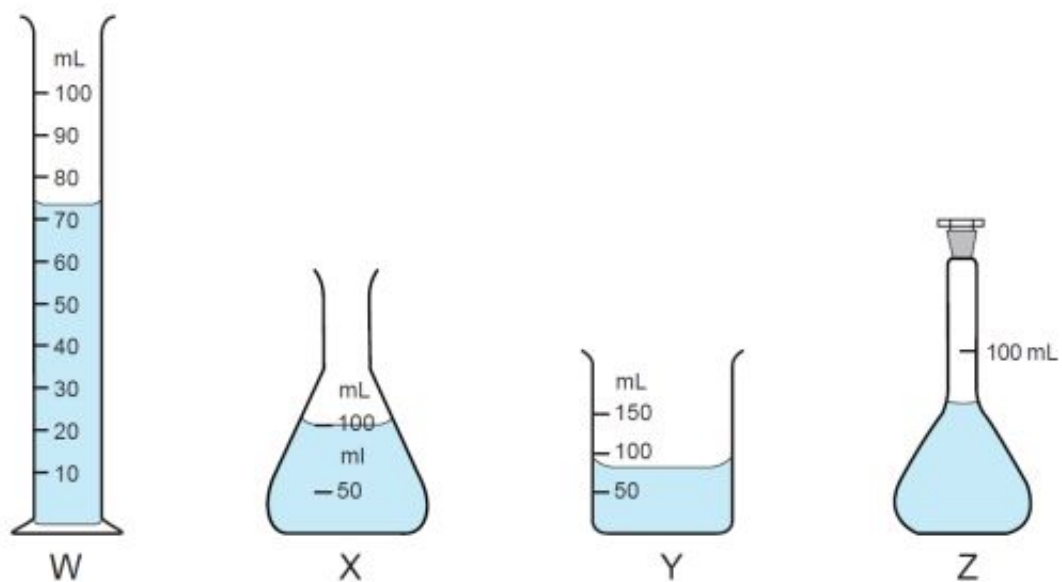
2. Solar panels convert energy from sunlight into electricity.



Which of the following would affect the amount of energy captured by the solar panels?

- (A) how hot the day is
- (B) how strong the roof is
- (C) how much energy is needed
- (D) how sunny it is during the day

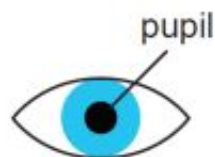
3. The diagram shows four containers, each holding some water.



Which container has the greatest amount of water?

- (A) W (B) X (C) Y (D) Z

4. The pupil is the opening in the centre of the eye. In bright light, the pupil becomes smaller so less light enters the eye. In the dark, the pupil becomes larger to let more light into the eye.



On a sunny afternoon, Simon walked into a dark cinema. Soon after, the movie started with a bright screen.

Which of these shows the changes in Simon's pupils?

- (A) Three eyes are shown in a row, connected by right-pointing arrows. The first eye has a small black pupil. The second eye has a large black pupil. The third eye has a small black pupil.
- (B) Three eyes are shown in a row, connected by right-pointing arrows. The first eye has a small black pupil. The second eye has a smaller black pupil. The third eye has an even smaller black pupil.
- (C) Three eyes are shown in a row, connected by right-pointing arrows. The first eye has a small black pupil. The second eye has a larger black pupil. The third eye has a medium-sized black pupil.
- (D) Three eyes are shown in a row, connected by right-pointing arrows. The first eye has a small black pupil. The second eye has a larger black pupil. The third eye has an even larger black pupil.

5. In the first photograph the arrow shows the length of a cicada's body.

In the second photograph the arrow shows the width of a cicada's thorax.



Body length



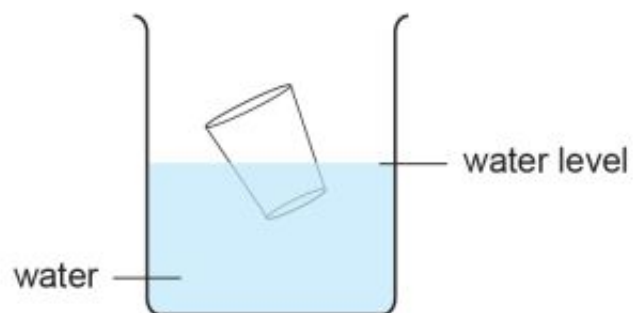
Thorax width

Which option has both the correct body length and correct thorax width?

	Body length (cm)	Thorax width (cm)
(A)	$4\frac{1}{2}$	$2\frac{1}{2}$
(B)	$4\frac{1}{2}$	2
(C)	$6\frac{1}{2}$	$2\frac{1}{2}$
(D)	$6\frac{1}{2}$	2

6. Lisa placed an empty plastic cup in a container of water.

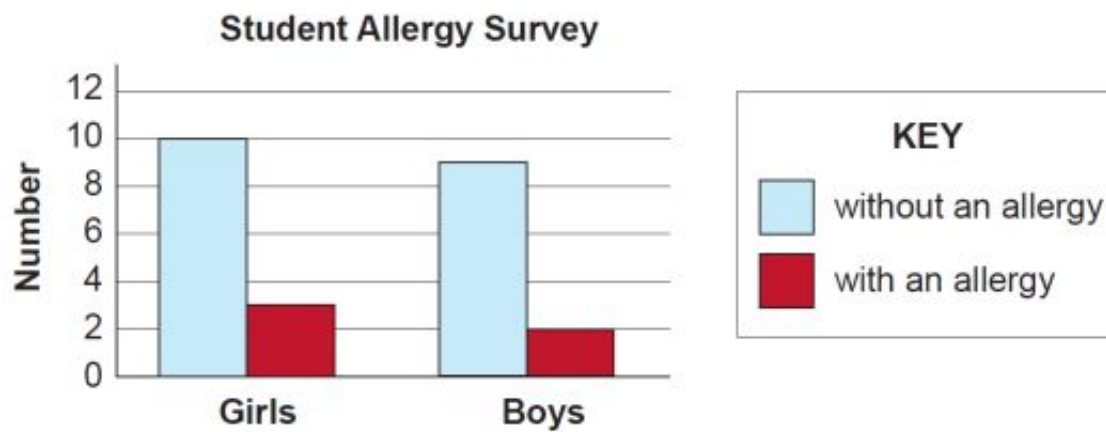
The diagram shows the plastic cup floating in the water.



What will happen if Lisa places a marble into the plastic cup?

- (A) The water level will move downwards.
- (B) The water level will move upwards.
- (C) The water level will not change.
- (D) The amount of water in the container will increase.

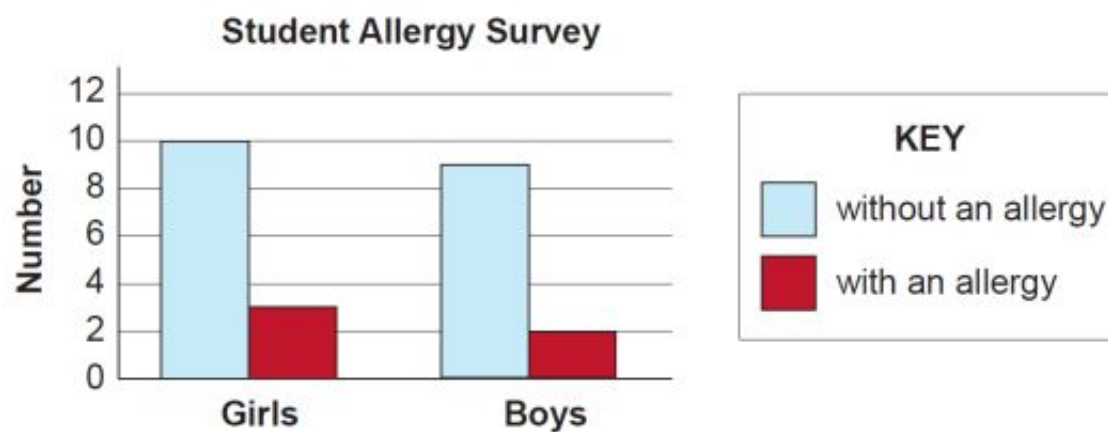
7. The graph shows the results of a student allergy survey in one class.



Which of the following groups has the greatest number of students in it?

- | | |
|------------------------------|------------------------------|
| (A) students with an allergy | (B) girls with an allergy |
| (C) boys without an allergy | (D) girls without an allergy |

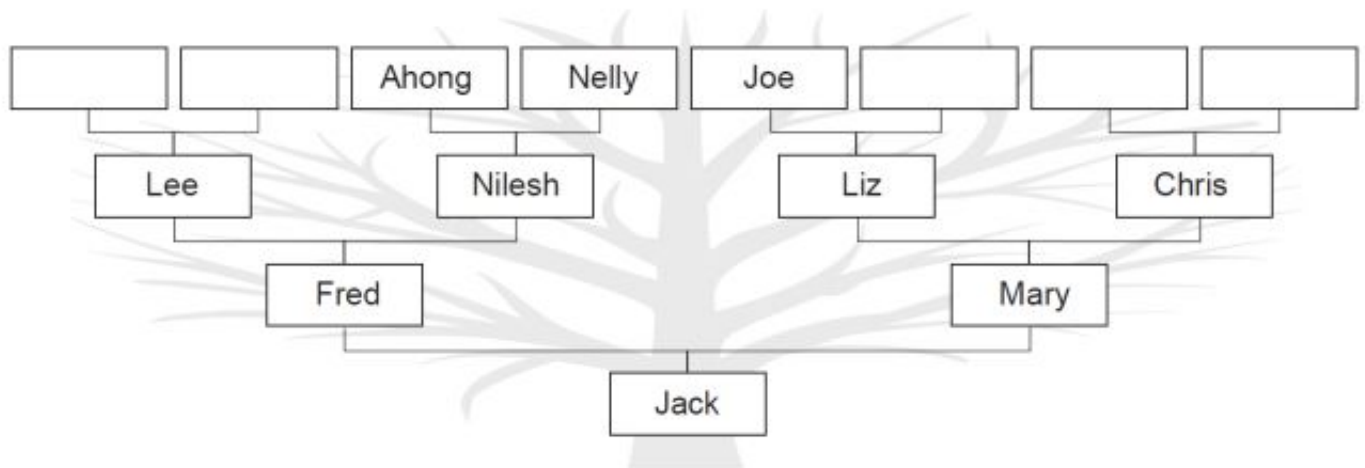
8. The graph shows the results of a student allergy survey in one class.



How many students are there in this class?

- (A) 24 (B) 22 (C) 19 (D) 10

9. Jack is drawing his family tree. The diagram shows what he has already drawn.

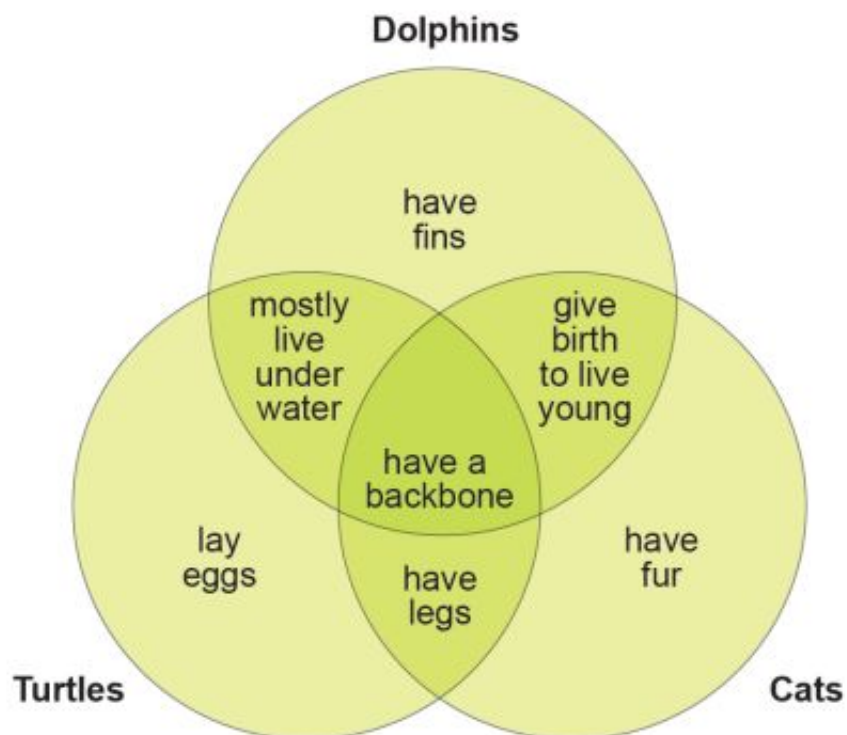


Liz and Chris are two of Jack's grandparents.

Who are two of Fred's grandparents?

- | | |
|---------------------|-------------------|
| (A) Lee and Niles | (B) Nelly and Joe |
| (C) Ahong and Nelly | (D) Liz and Chris |

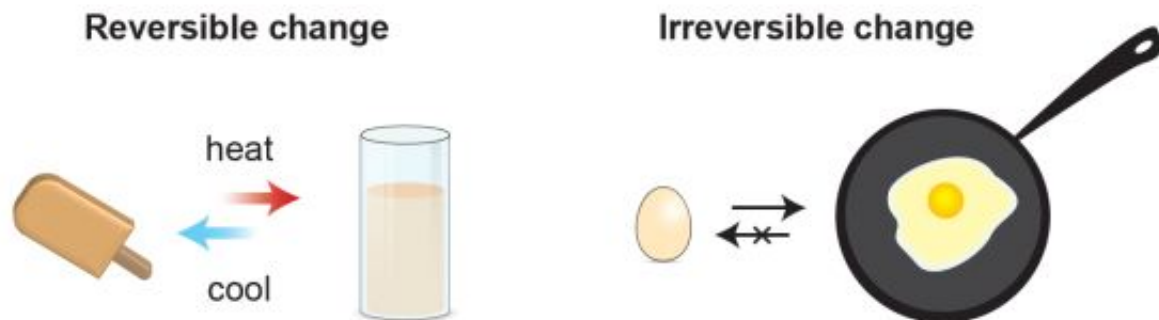
10. The diagram shows some features of three animals.



Which feature is common to dolphins and cats but not to turtles?

- | | |
|------------------------------------|-----------------------------------|
| (A) They lay eggs. | (B) They have a backbone. |
| (C) They give birth to live young. | (D) They mostly live under water. |

11. Some changes can be easily undone. These changes are called reversible changes. Changes that cannot be undone are called irreversible changes.

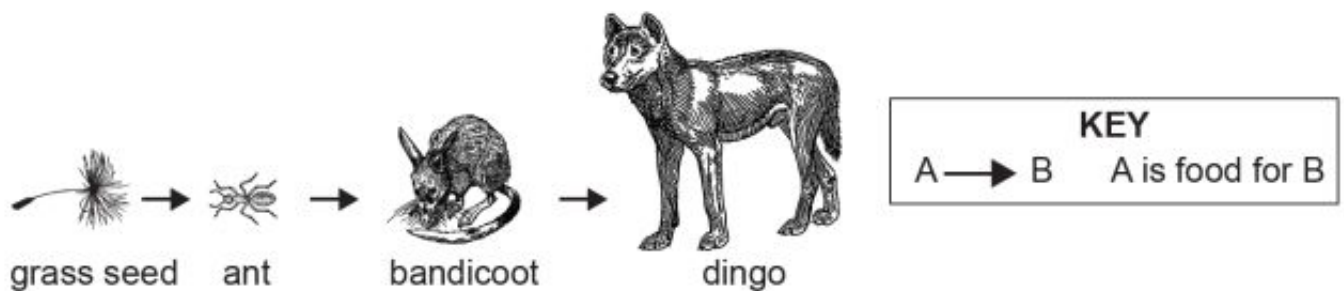


Which of the following is a reversible change?

- (A) green leaves turning yellow in autumn
- (B) a piece of metal being pressed into coins
- (C) a piece of apple going brown
- (D) paper burning to ashes

12. A food chain shows how nutrients and energy in food are passed on from one living thing to the next.

The diagram shows part of a food chain.



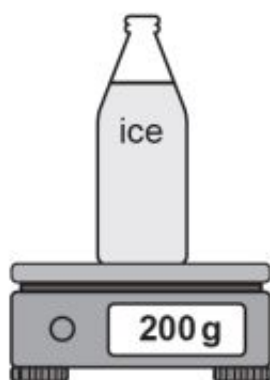
Due to the spread of a disease, the number of bandicoots decreased rapidly.

What is likely to happen as a result?

- (A) The number of dingoes and the number of ants will both increase.
- (B) The number of dingoes will decrease and the number of ants will increase.
- (C) The number of dingoes will increase and the number of ants will decrease.
- (D) The number of dingoes and the number of ants will both decrease.

13. When water freezes, it takes up more space.

Anika weighed a bottle of frozen water on a balance, making sure the outside of the bottle was wiped dry.



She then left it on the table overnight.

Which option shows the mass and the level of water in the bottle when all the ice had melted?

	Mass (g)	Level of water
(A)	200	lower
(B)	more than 200	lower
(C)	less than 200	higher
(D)	200	higher

14. To conduct a fair test, the investigators must:

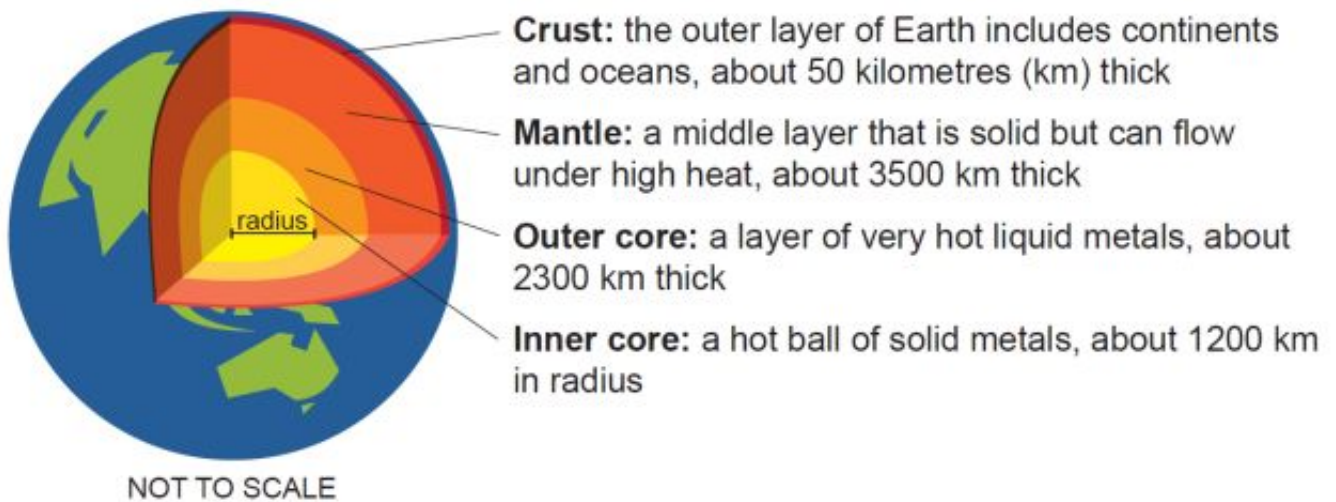
- **change** only one thing for each test
- **keep** everything else the same, and
- **observe or measure** one change that happens.

A group of Year 3 students want to see if the shadow of a stick changes length during the day.

What must the students change, what must they keep the same, and what must they measure to make their investigation a fair test?

	Change	Keep the same	Measure
(A)	the time of the day	the length of the shadow	the length of the stick
(B)	the length of the shadow	the length of the stick	the time of the day
(C)	the time of the day	the length of the stick	the length of the shadow
(D)	the length of the stick	the time of the day	the length of the shadow

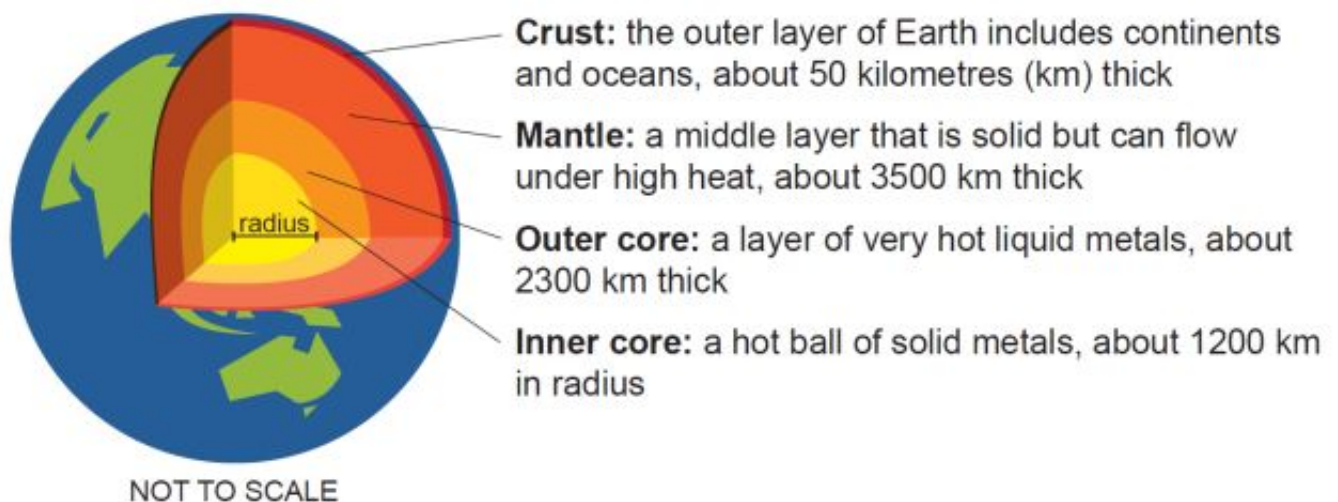
15. Earth is made up of four layers.



Which layer is the thickest?

- (A) crust (B) mantle (C) outer core (D) inner core

16. Earth is made up of four layers.



Which statement is supported by the information?

- (A) The mantle contains ocean floors.
- (B) The inner core is a ball of liquid.
- (C) The outer core is a solid layer surrounding the inner core.
- (D) The inside of Earth consists of mostly hot materials that flow.

17. Animals that have a backbone are called vertebrates.

The table shows five groups of vertebrates and their characteristics.

	Fish	Amphibians	Reptiles	Birds	Mammals
Form of reproduction	lay eggs with no outer shell	lay eggs with no outer shell	lay eggs with outer shell	lay eggs with outer shell	give birth to live young
Body covering	interlocking scales	moist skin	interlocking scales	feathers	dry skin with hair
Method of breathing	through gills	through skin or gills; lungs as adult	through lungs	through lungs	through lungs
Cold- or warm-blooded	cold-blooded	cold-blooded	cold-blooded	warm-blooded	warm-blooded

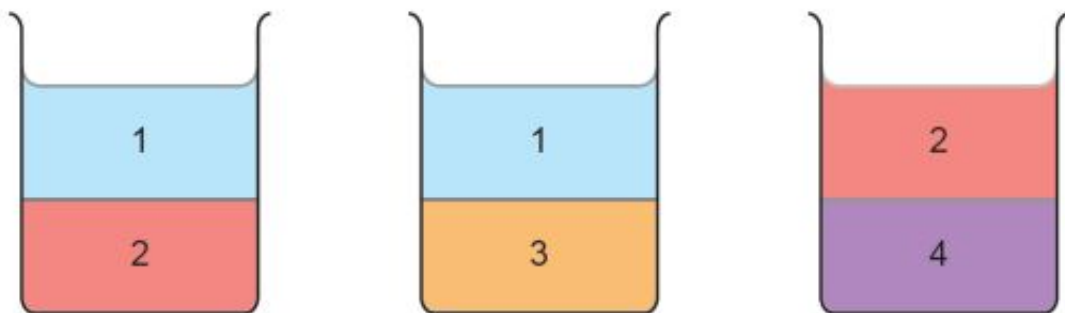
Tuatara are animals that have interlocking scales and they breathe through lungs.

Which statement about tuatara is true?

- (A) They lay eggs with no outer shell.
- (B) They are fish.
- (C) They are reptiles.
- (D) They are warm-blooded.

18. Lily has four different liquids that do not mix with each other. She tested three different combinations of these liquids.

The diagrams show her results.

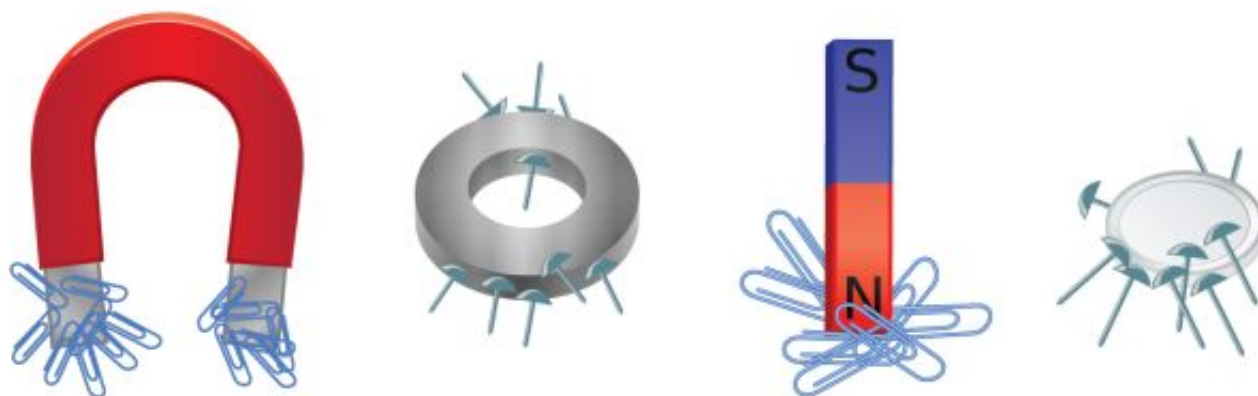


Which statement is supported by the diagrams?

- | | |
|--------------------------------------|--------------------------------------|
| (A) Liquid 1 will float on liquid 4. | (B) Liquid 2 will float on liquid 3. |
| (C) Liquid 3 will float on liquid 2. | (D) Liquid 3 will float on liquid 4. |

19. Magnets can attract objects made of iron.

Ying wants to compare the strengths of four magnets. She used some paper clips and thumb tacks to carry out her investigation. This shows her results.



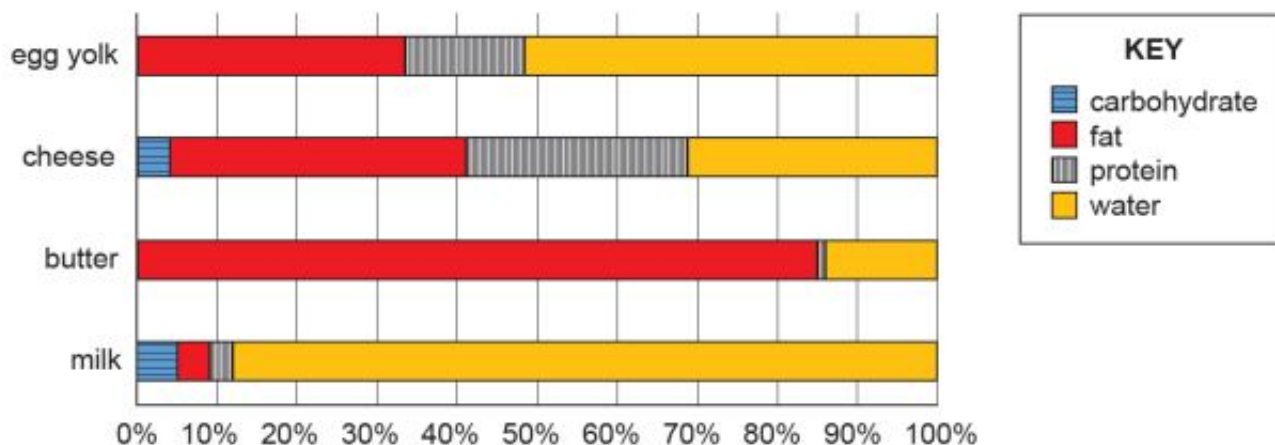
This investigation is not a fair test.

What can Ying do to make it a fair test?

- (A) use fewer paper clips
- (B) use the same size paper clips for each magnet
- (C) use the same number of paper clips and thumb tacks for each magnet
- (D) use large paper clips for the large magnet and small paper clips for the small magnet

20. Eggs and dairy products are made of four major components: carbohydrate, fat, protein and water.

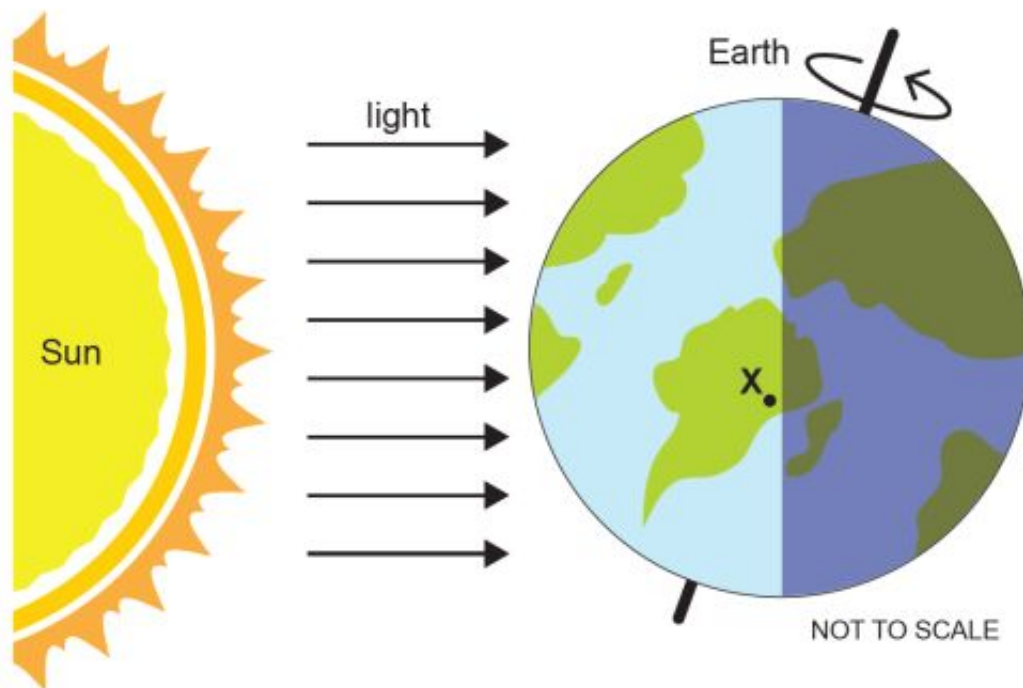
The graph shows the composition of four types of food.



For 100 g of each food, which food contains the least amount of water?

- (A) egg yolk
- (B) cheese
- (C) butter
- (D) milk

21. As Earth orbits the Sun, it also spins on its axis. It takes about 24 hours for one complete spin. It is daytime on the side of Earth facing the Sun and it is night-time on the side facing away from the Sun.

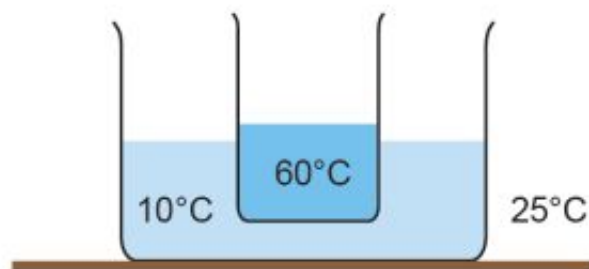


What is the time of day at the location marked X on the diagram?

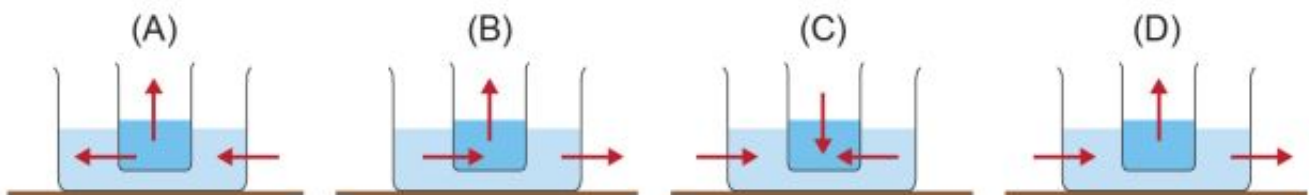
- (A) midday (B) midnight (C) sunrise (D) sunset

22. When two objects are in contact with one another, heat flows from the object with the higher temperature to the object with the lower temperature.

The diagram shows a small beaker of water placed in a larger container of water on a table. It also shows the temperature of the water in each container and the air temperature.



Which diagram correctly shows the direction in which heat flows at this time?



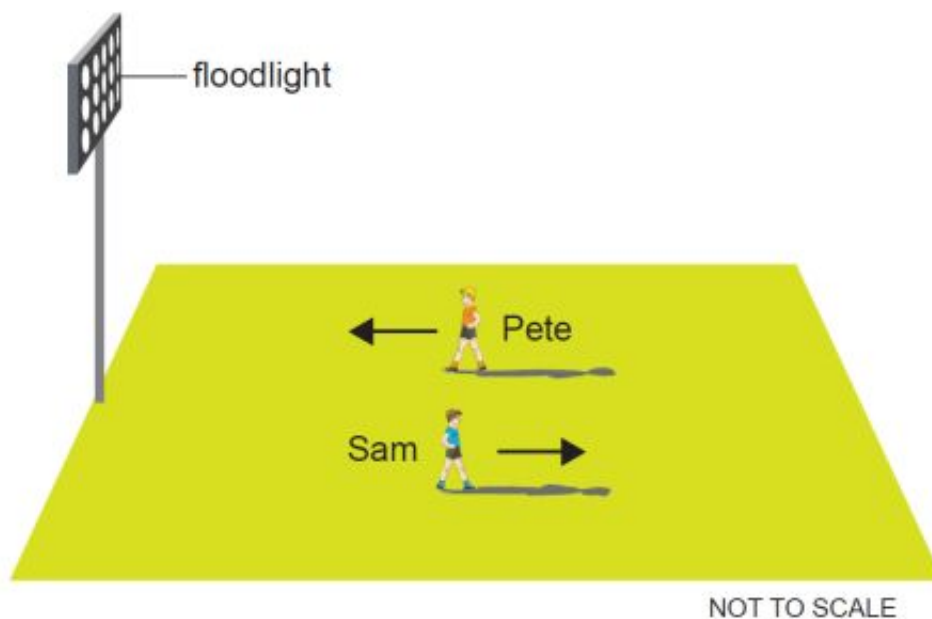
23. This table lists the maximum length and mass of different types of adult whales.

Type of whale	Maximum length of adult (metres)	Maximum mass of adult (tonnes)
Arnoux's beaked	9	8
Blue	30	150
Fin	27	90
Humpback	16	48
Minke	10	10
Southern bottlenose	9	4.5
Southern right	17	100
Sperm	15	40

Which two types of whales have the closest maximum mass?

- (A) Arnoux's beaked and Southern bottlenose (B) Humpback and Sperm
(C) Blue and Southern right (D) Arnoux's beaked and Minke

24. On a football field, Pete and Sam cast shadows in the floodlight. They start walking in opposite directions, as shown.

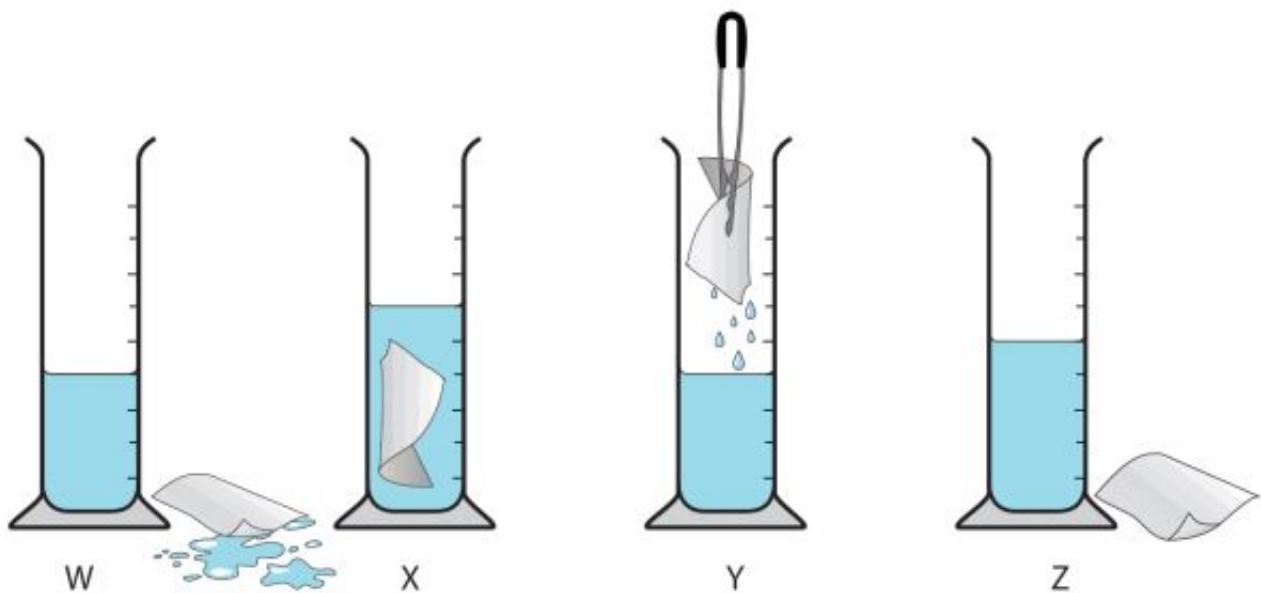


What will happen to their shadows as they walk?

- (A) Both shadows become longer.
- (B) Sam's shadow becomes longer and Pete's shadow becomes shorter.
- (C) Both shadows become shorter.
- (D) Pete's shadow becomes longer and Sam's shadow becomes shorter.

25. Laura carried out an investigation to find out how much water a paper towel could absorb.

The diagram shows four steps of her investigation, but not in the correct order.



Which of these lists the steps in the correct order?

- (A) W, Y, X, Z (B) Z, W, X, Y (C) Y, X, W, Z (D) Z, X, Y, W

26. Mrs Black showed this science experiment to her Year 3 class.

She put a small drop of four food colours onto the surface of some milk in a dish.

After 1 minute she carefully added a drop of detergent onto the middle of the dish. The colours immediately started moving and mixing with each other forming colourful patterns on the surface of the milk.

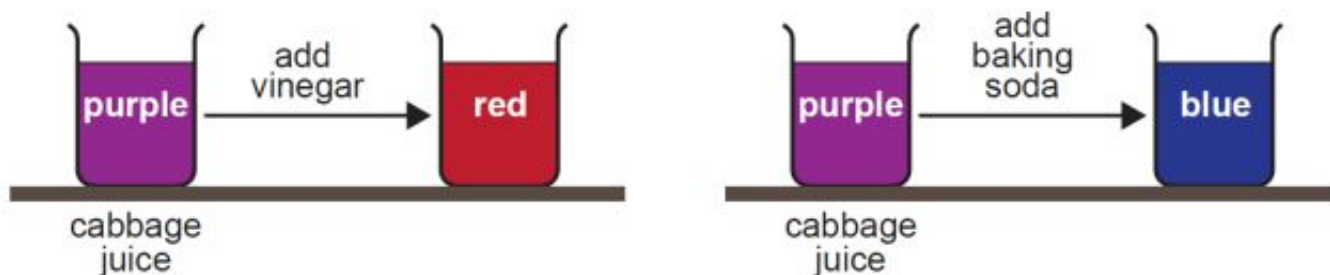


Which statement is supported by these observations?

- (A) The milk surface is constantly moving, making the colours mix.
- (B) The milk surface is smooth, making the colours slide around.
- (C) The colours mix with the detergent, making the milk surface move and mix.
- (D) The detergent mixes with milk, making the milk surface move and the colours mix.

27. Purple cabbage juice can be used to indicate whether a substance is alkaline or acidic.

Max took two cups of purple cabbage juice. He then added vinegar to one and baking soda to the other and noted the colour changes.



Max knew that vinegar is acidic and baking soda is alkaline. Max added some liquid he found in the kitchen to a new cup of cabbage juice. The cabbage juice turned red.

What can Max conclude from his experiment?

- (A) The cabbage juice was acidic.
- (B) The liquid contained vinegar.
- (C) The liquid was acidic.
- (D) The liquid was red.

28. Gears are wheels with teeth along their edge. The teeth on two gears can mesh so that a turning gear can make another gear turn without slipping.

This diagram shows three gears.



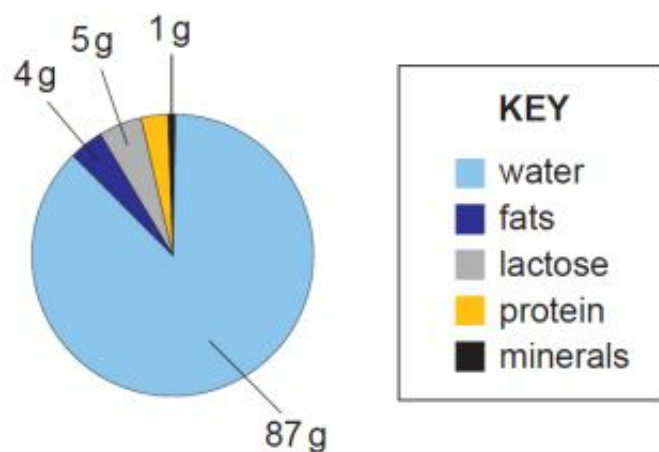
Gear X is turning anticlockwise.

How does gear Z move compared with gear X?

- (A) It turns faster, in a clockwise direction.
- (B) It turns slower, in a clockwise direction.
- (C) It turns slower, in an anticlockwise direction.
- (D) It turns faster, in an anticlockwise direction.

29. Milk is made up of water and some solids. The solids include fats and non-fat solids.

The sector graph shows what is in 100 grams (g) of cow's milk.



Based on the diagram, how many grams of non-fat solids are there in 100 g of cow's milk?

- (A) 3 (B) 9 (C) 10 (D) 13

30. The table lists the times of sunrise and sunset in Sydney over seven days during May 2017.

Date	Sunrise (am)	Sunset (pm)
May 21	6:44	4:58
May 22	6:45	4:58
May 23	6:45	4:57
May 24	6:46	4:57
May 25	6:47	4:56
May 26	6:47	4:56
May 27	6:48	4:55

Based on the table, what were the most likely times for sunrise and sunset on May 28, 2017?

	Sunrise (am)	Sunset (pm)
(A)	6:49	4:55
(B)	6:49	4:56
(C)	6:48	4:56
(D)	6:47	4:55

Question Number	Area Assessed	Description	Correct Answer
1	Observing/Measuring	Recognise observable properties of a living thing	C
2	Predicting/Concluding	Identify a cause for an effect	D
3	Observing/Measuring	Compare the volume of liquid in different containers	B
4	Interpreting	Sequence changes based on given information	A
5	Observing/Measuring	Measure lengths using a ruler	B
6	Predicting/Concluding	Predict the outcome of an action	B
7	Interpreting	Interpret a column graph	D
8	Interpreting	Calculate a total using data from a column graph	A
9	Interpreting	Interpret relationships in a diagram of a family tree	C
10	Predicting/Concluding	Draw a conclusion from a Venn diagram	C

11	Predicting/Concluding	Differentiate the types of changes	B
12	Reasoning/Problem solving	Predict a likely outcome following a change in a simple food chain	B
13	Predicting/Concluding	Predict the result of an investigation	A
14	Investigating	Identify three types of variables in a simple investigation	C
15	Observing/Measuring	Compare a property using information in a diagram	B
16	Predicting/Concluding	Draw a conclusion based on information in a diagram	D
17	Predicting/Concluding	Draw a conclusion from information in a table	C
18	Reasoning/Problem solving	Solve a problem using relative densities of liquids	A
19	Investigating	Recognise the conditions for a fair test in a given investigation	B
20	Interpreting	Interpret a composite column graph	C

21	Interpreting	Interpret an Earth-Sun diagram	D
22	Predicting/Concluding	Predict the result of a given scenario	A
23	Interpreting	Identify specific information from tabulated data	D
24	Predicting/Concluding	Predict a causal effect	B
25	Investigating	Sequence the steps of an investigation using diagrams	D
26	Investigating	Draw a conclusion based on the observations in an investigation	D
27	Investigating	Draw a conclusion based on the results of an investigation	C
28	Reasoning/Problem solving	Solve a problem involving gears	C
29	Reasoning/Problem solving	Determine a value using information in a sector graph	B
30	Reasoning/Problem solving	Identify patterns in tabulated data and make a prediction	A